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Control device for aerosol inhalation device and aerosol inhalation device

Abstract

Control device for aerosol inhalation device, includes operational amplifier including output terminal configured to generate voltage according to voltage applied to load configured to heat aerosol source and having correlation between temperature and electrical resistance value, control unit including input terminal and configured to perform processing based on voltage applied to the input terminal, and voltage dividing circuit configured to electrically connect the output terminal of the operational amplifier and the input terminal of the control unit. Power supply voltage of the operational amplifier is higher than power supply voltage of the control unit, and equals voltage applied to aerosol generation circuit including the load, and one of inverting input terminal and noninverting input terminal of the operational amplifier is electrically connected to the aerosol generation circuit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/570,402, filed on Jan. 7, 2022, which is a continuation of U.S. application Ser. No. 16/885,337, filed on May 28, 2020 (now U.S. Pat. No. 11,490,661), which claims priority to and the benefit of Japanese Patent Application No. 2019-102263, filed on May 31, 2019, the entire disclosure of each are incorporated herein by reference. (2) This application is also related to U.S. patent application Ser. No. 16/885,343 (14598US01), entitled “AEROSOL INHALATION DEVICE AND CONTROL DEVICE FOR AEROSOL INHALATION DEVICE” and U.S. patent application Ser. No. 16/885,393 (14600US01), entitled “CONTROL DEVICE FOR AEROSOL INHALATION DEVICE AND AEROSOL INHALATION DEVICE”, all filed on the same day as this application, and all hereby incorporated by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a control device for an aerosol inhalation device configured to generate an aerosol to be inhaled by a user and the aerosol suction device. Note that the aerosol inhalation device is also called an aerosol generation device.

Description of the Related Art

(2) In a general aerosol inhalation device such as an electronic cigarette, a heated tobacco product, or a nebulizer, which is used to generate an aerosol to be inhaled by a user, if the user performs inhalation when an aerosol source (to be also referred to as an “aerosol forming substrate” hereinafter) that changes to an aerosol by atomization is in shortage, a sufficient aerosol cannot be supplied to the user. Additionally, in a case of an electronic cigarette or a heated tobacco product, it is impossible to generate an aerosol having an intended flavor.

(3) To detect the remaining amount of an aerosol source, there exists a method using a heater having a PTC characteristic that changes an electrical resistance value in accordance with a temperature. In this method, based on the electrical resistance value of a heater, which is obtained from a voltage applied to the heater or a voltage according to the voltage (to be referred to as “a voltage applied to a heater, or the like” hereinafter), it can be determined whether the aerosol source is in shortage.

(4) There also exists an aerosol inhalation device configured to exchange a cartridge including an aerosol source and a heater that heats the aerosol source when the aerosol source has run out. The exchange can also be detected by the method using the voltage applied to the heater, or the like.

(5) That is, in the aerosol inhalation device, processing based on the voltage applied to the heater, or the like is sometimes executed. To execute the processing based on the voltage applied to the heater, or the like, a control device for the aerosol inhalation device can include an operational amplifier configured to output a voltage according to the voltage applied to the heater, and a control unit such as a microcontroller unit (MCU) to which a voltage according to the voltage is applied. In such a control device, the operational amplifier or the control unit may not normally operate due to an excessive voltage (to be referred to as an “overvoltage” hereinafter) if the operational amplifier or the control unit is not protected from the relatively high voltage applied to the heater.

(6) PTL 1 discloses a circuit arrangement used to control the coil temperature of a resistance heating element of a vaporizing device. This circuit arrangement includes a battery BATT (23), a resistor R_COIL (74) of a coil to which a voltage from the battery BATT (23) can be applied, and an MCU (72) that is connected to the resistor R_COIL (74) of the coil and uses the voltage of the battery BATT (23) as a power supply voltage (see paragraph 0191 and FIG. 17B and the like).

However, PTL 1 neither discloses nor suggests an arrangement that prevents an overvoltage from being applied to an operational amplifier or the MCU (72).

(7) PTL 2 discloses a circuit arrangement for measuring the resistance of a heating element of an aerosol generation system. This circuit arrangement includes a heater 501, and a microprocessor 507 connected to the heater 501, to which a voltage V1 can be applied (see paragraph 0081 and FIG. 5 and the like). However, PTL 2 neither discloses nor suggests an arrangement that prevents an overvoltage from being applied to an operational amplifier or the microprocessor 507.

(8) PTL 3 discloses a principle block diagram of a battery assembly of an electronic cigarette. This block diagram includes a battery assembly 101, a spraying assembly 20 connected to the battery assembly 101, and a microcontroller 102 connected to the battery assembly 101 and the spraying assembly 20 (see paragraph 0021 and FIG. 2 and the like). However, PTL 3 neither discloses nor suggests an arrangement that prevents an overvoltage from being applied to an operational amplifier or the microcontroller 102.

(9) PTL 4 discloses a control body 102 and a cartridge 104 of an aerosol sending device. The cartridge 104 includes a heater 222, and the control body 102 includes a microprocessor 310 which is electrically connected to the heater 222 when the cartridge 104 is connected, and to which a voltage V_{is} is applied (see paragraph 0064 and FIG. 3 and the like). However, PTL 4 neither discloses nor suggests an arrangement that prevents an overvoltage from being applied to an operational amplifier or the microprocessor 310.

PATENT LITERATURES

(10) PTL 1: Japanese PCT National Publication No. 2017-501805 PTL 2: Japanese PCT National Publication No. 2014-501105 PTL 3: Japanese PCT National Publication No. 2017-503520 PTL 4: Japanese PCT National Publication No. 2018-526983

SUMMARY OF THE INVENTION

(11) The present invention has been made in consideration of the above-described points.

(12) The first problem to be solved by the present invention is to provide a control device for an aerosol inhalation device, which includes an operational amplifier and the like, in which the voltage of a circuit electrically connected to the noninverting input terminal or the inverting input terminal of the operational amplifier is relatively high, and which can prevent an overvoltage from being applied to the operational amplifier or the like.

(13) The second problem to be solved by the present invention is to provide a control device for an aerosol inhalation device, which includes a control unit electrically connected to the output terminal of an operational amplifier, and can prevent an overvoltage higher than the power supply voltage of the control unit from being applied to the control unit.

(14) In order to solve the above-described first problem, according to the embodiment of the present invention, there is provided a control device for an aerosol inhalation device, comprising an operational amplifier configured to perform output according to a voltage applied to a load configured to heat an aerosol source and having a correlation between a temperature and an electrical resistance value, a control unit configured to perform processing based on the voltage according to the output, and a diode having an anode electrically connected to one of an inverting input terminal and a noninverting input terminal of the operational amplifier.

(15) The control device for the aerosol inhalation device according to the embodiment may further comprise a circuit configured to electrically connect a power supply and the load, the circuit may be formed by a first region, and a second region in which a maximum voltage is lower than a maximum voltage in the first region, or an applied voltage is lower than a voltage applied to the first region, and of the inverting input terminal and the noninverting input terminal, a terminal to which the anode of the diode is electrically connected may electrically be connected to the first region.

(16) In the embodiment, the anode of the diode may electrically be connected to the first region.

(17) According to this arrangement, a forward current flows to the diode, thereby preventing an

overvoltage from being applied to the input terminal of the operational amplifier. Hence, it is possible to prevent an operation error of the operational amplifier or the like, which is caused by application of an overvoltage to the input terminal.

(18) In the embodiment, the load may electrically be connected to or included in the first region.

(19) According to the arrangement, a relatively high voltage can be applied to the load, for example, a heater, and an aerosol can be generated more efficiently.

(20) The control device for the aerosol inhalation device according to the embodiment may further comprise a converter configured to output a predetermined voltage to be applied to the first region.

(21) In the embodiment, the converter may comprise a switching regulator.

(22) According to the arrangement, the voltage can stably be applied to the load, for example, a heater independently of the remaining amount or degradation state of the power supply, and therefore, aerosol generation can be controlled more stably.

(23) In the embodiment, the converter may further be configured to output a voltage that prevents a forward current flowing to the diode from exceeding an allowable value and allows the load to generate an aerosol.

(24) According to the arrangement, an overcurrent can be prevented from flowing to the diode, and an aerosol can be generated. It is therefore possible to simultaneously solve conflicting problems, that is, generation of a sufficient aerosol and protection of the diode.

(25) In the embodiment, the converter may further be configured to be able to output a plurality of voltages or voltages in a range, the voltages preventing a forward current flowing to the diode from exceeding an allowable value and allowing the load to generate an aerosol.

(26) In the embodiment, the control unit may be configured to adjust an output voltage of the converter in the plurality of voltages or the voltages in the range in accordance with a type of the load.

(27) According to the arrangement, an appropriate voltage can be used in accordance with the type of the load, and it is therefore possible to implement an aerosol inhalation device capable of using various kinds of loads using a single control device.

(28) In the embodiment, a power supply terminal of the operational amplifier may electrically be connected to the second region.

(29) According to the arrangement, a forward current flows to the diode, thereby preventing an overvoltage from being applied to the input terminal of the operational amplifier. Hence, it is possible to prevent an operation error of the operational amplifier or the like, which is caused by application of an overvoltage to the input terminal.

(30) The control device for the aerosol inhalation device according to the embodiment may further comprise a regulator configured to output a voltage to be applied to the second region, and the control unit may be configured such that a power supply terminal of the control unit is electrically connected to the second region.

(31) According to the arrangement, an appropriate voltage can be applied by the regulator to the control unit that is driven by a relatively low voltage. At the same time, since the voltage in the second region is stabilized by the regulator, power can be released to the second region via the diode at the time of overvoltage generation in the inverting input terminal or the noninverting input terminal of the operational amplifier.

(32) In the embodiment, a cathode of the diode may electrically be connected to the second region.

(33) In the embodiment, an electrical resistance value of the diode in a forward direction may be smaller than an electrical resistance value of the operational amplifier.

(34) In the embodiment, a cathode of the diode may electrically be connected to a power supply terminal of the operational amplifier, or a potential at the cathode of the diode may equal a potential at the power supply terminal of the operational amplifier.

(35) According to the arrangement, a forward current flows to the diode, thereby preventing an overvoltage from being applied to the input terminal of the operational amplifier. Hence, it is

possible to prevent an operation error of the operational amplifier or the like, which is caused by application of an overvoltage to the input terminal.

(36) The control device for the aerosol inhalation device according to the embodiment further comprises a first circuit and a second circuit, which are electrically connected in parallel between a power supply and the load, the first circuit and the second circuit including a first switch and a second switch, respectively, and being configured such that an electrical resistance value of the second circuit is higher than an electrical resistance value of the first circuit, and the control unit may further be configured to acquire the voltage according to the output of the operational amplifier during a time when the second switch is in an ON state.

(37) According to the arrangement, since a dedicated circuit used to acquire a voltage, which includes a known resistor, is provided, the acquisition accuracy is improved by the known resistor. Simultaneously, it is possible to generate an aerosol without any influence of the known resistor. This improves the use efficiency of the storage capacity of the power supply, for example, a lithium ion secondary battery.

(38) In the embodiment, the control unit may be configured to set the first switch in the ON state to generate an aerosol.

(39) According to the arrangement, power can be supplied to the load without any influence of the electrical resistance value of the second circuit, and it is therefore possible to efficiently generate an aerosol. In other words, a large amount of aerosol can be generated by one charge.

(40) The control device for the aerosol inhalation device according to the embodiment further comprises the load electrically connected in series with the first circuit and the second circuit, and a second resistor electrically connected in series with the first circuit and the second circuit and electrically connected in parallel with the load, and the load may be configured to be detachable from the first circuit, the second circuit, and the second resistor.

(41) In the embodiment, the second circuit may include a first resistor, and an electrical resistance value of the first resistor and an electrical resistance value of the second resistor may equal.

(42) According to the arrangement, even if the load is detached, an electrical path via the second resistor exists. Hence, since the operational amplifier can perform an output according to the voltage applied to the second resistor, the output is stable.

(43) Also, in order to solve the above-described first problem, according to the embodiment of the present invention, there is provided a control device for an aerosol inhalation device, comprising a comparator configured to perform output according to a voltage applied to a load configured to heat an aerosol source and having a correlation between a temperature and an electrical resistance value, a control unit configured to perform processing based on the voltage according to the output, and a diode having an anode electrically connected to an input terminal of the comparator.

(44) According to the arrangement, a forward current flows to the diode, thereby preventing an overvoltage from being applied to the input terminal of the comparator. Hence, it is possible to prevent an operation error of the comparator or the like, which is caused by application of an overvoltage to the input terminal.

(45) Furthermore, in order to solve the above-described first problem, according to the embodiment of the present invention, there is provided a control device for an aerosol inhalation device, comprising an operational amplifier configured to perform output according to a voltage applied to a load configured to heat an aerosol source and having a correlation between a temperature and an electrical resistance value, a control unit configured to perform processing based on the voltage according to the output, and an aerosol generation circuit including the load, wherein a power supply voltage of the operational amplifier equals a voltage applied to the aerosol generation circuit, and one of an inverting input terminal and a noninverting input terminal of the operational amplifier is electrically connected to the aerosol generation circuit.

(46) According to the arrangement, since the voltage applied to the inverting input terminal or the noninverting input terminal of the operational amplifier is equal to or lower than the power supply

voltage of the operational amplifier. Hence, it is possible to prevent an operation error of the operational amplifier or the like, which is caused by application of an overvoltage to the operational amplifier.

(47) Furthermore, in order to solve the above-described first problem, according to the embodiment of the present invention, there is provided an aerosol inhalation device comprising a control device as described above.

(48) According to the arrangement, a forward current flows to the diode, thereby preventing an overvoltage from being applied to the input terminal of the operational amplifier or the comparator. Hence, it is possible to prevent an operation error of the operational amplifier or the comparator or the like, which is caused by application of an overvoltage to the input terminal.

(49) Alternatively, according to the arrangement, since the voltage applied to the inverting input terminal or the noninverting input terminal of the operational amplifier is equal to or lower than the power supply voltage of the operational amplifier. Hence, it is possible to prevent an operation error of the operational amplifier or the like, which is caused by application of an overvoltage to the operational amplifier.

(50) In order to solve the above-described second problem, according to the embodiment of the present invention, there is provided a control device for an aerosol inhalation device, comprising an operational amplifier including an output terminal configured to generate a voltage according to a voltage applied to a load configured to heat an aerosol source and having a correlation between a temperature and an electrical resistance value, a control unit including an input terminal and configured to perform processing based on a voltage applied to the input terminal, and a voltage dividing circuit configured to electrically connect the output terminal of the operational amplifier and the input terminal of the control unit.

(51) In the embodiment, the voltage dividing circuit may be configured such that the voltage applied to the input terminal of the control unit becomes not more than a power supply voltage of the control unit.

(52) In the embodiment, a power supply voltage of the operational amplifier may be higher than the power supply voltage of the control unit.

(53) According to the arrangement, a voltage stepped down by the voltage dividing circuit is applied to the input terminal of the control unit. This can prevent an operation error of the control unit or the like, which is caused by application of an overvoltage to the input terminal.

(54) In the embodiment, an amplification factor of the operational amplifier may be set such that the voltage applied to the input terminal of the control unit becomes not more than the power supply voltage of the control unit.

(55) According to the arrangement, an overvoltage is not applied to the input terminal of the control unit. This can prevent an operation error of the control unit or the like, which is caused by application of an overvoltage to the input terminal.

(56) In the embodiment, a power supply voltage of the operational amplifier may equal a voltage applied to an aerosol generation circuit including the load, and one of an inverting input terminal and a noninverting input terminal of the operational amplifier may electrically be connected to the aerosol generation circuit.

(57) According to the arrangement, a voltage higher than the power supply voltage of the operational amplifier is not applied to the input terminal of the operational amplifier. Hence, it is possible to prevent an operation error of the operational amplifier or the like, which is caused by application of an overvoltage to the input terminal.

(58) The control device for the aerosol inhalation device according to the embodiment may further comprise a converter configured to apply a predetermined voltage to an aerosol generation circuit including the load.

(59) In the embodiment, an output terminal of the converter may electrically be connected to the aerosol generation circuit and a power supply terminal of the operational amplifier, and one of an

inverting input terminal and a noninverting input terminal of the operational amplifier may electrically be connected to the aerosol generation circuit.

(60) According to the arrangement, the voltage can stably be applied to the load, for example, a heater, and therefore, aerosol generation can be controlled more accurately.

(61) In the embodiment, the converter may be configured to output a voltage that makes the voltage applied to the input terminal of the control unit not more than a power supply voltage of the control unit and allows the load to generate an aerosol.

(62) In the embodiment, the voltage dividing circuit may be configured such that the voltage applied to the input terminal of the control unit becomes not more than the power supply voltage of the control unit.

(63) According to the arrangement, it is possible to prevent an overvoltage from being applied to the input terminal of the control unit and also generate an aerosol. It is therefore possible to prevent an operation error of the control unit or the like during aerosol generation.

(64) In the embodiment, the converter may comprise a switching regulator.

(65) According to the arrangement, the voltage can stably be applied to the load, for example, a heater, and therefore, aerosol generation can be controlled more accurately.

(66) In the embodiment, the converter may be configured to be able to output a plurality of voltages or voltages in a range, the voltages making the voltage applied to the input terminal of the control unit not more than a power supply voltage of the control unit and allowing the load to generate an aerosol.

(67) In the embodiment, the control unit may be configured to adjust an output voltage of the converter in accordance with a type of the load.

(68) According to the arrangement, an appropriate voltage can be used in accordance with the type of the load, and it is therefore possible to implement an aerosol inhalation device capable of using various kinds of loads using a single control device.

(69) The control device for the aerosol inhalation device according to the embodiment further comprises a first circuit and a second circuit, which are electrically connected in parallel between a power supply and the load, the first circuit and the second circuit including a first switch and a second switch, respectively, and being configured such that an electrical resistance value of the second circuit is larger than an electrical resistance value of the first circuit, and the control unit may further be configured to acquire the voltage applied to the input terminal during a time when the second switch is in an ON state.

(70) According to the arrangement, since a dedicated circuit used to acquire a voltage, which includes a known resistor, is provided, the acquisition accuracy is improved by the known resistor. Simultaneously, it is possible to generate an aerosol without any influence of the known resistor. This improves the use efficiency of the storage capacity of the power supply, for example, a lithium ion secondary battery.

(71) In the embodiment, the control unit may be configured to set the first switch in the ON state to generate an aerosol.

(72) According to the arrangement, power can be supplied to the load without any influence of the electrical resistance value of the second circuit, and it is therefore possible to efficiently generate an aerosol.

(73) The control device for the aerosol inhalation device according to the embodiment may further comprise the load electrically connected in series with the first circuit and the second circuit, and a second resistor electrically connected in series with the first circuit and the second circuit and electrically connected in parallel with the load, and the load may be configured to be detachable from the first circuit, the second circuit, and the second resistor.

(74) In the embodiment, the second circuit may include a first resistor, and an electrical resistance value of the first resistor and an electrical resistance value of the second resistor may equal.

(75) According to the arrangement, even if the load is detached, an electrical path via the second

resistor exists. Hence, since the operational amplifier can perform an output according to the voltage applied to the second resistor, the output is stable.

(76) Furthermore, in order to solve the above-described second problem, according to the embodiment of the present invention, there is provided an aerosol inhalation device comprising a control device as described above.

(77) According to the arrangement, a voltage stepped down by the voltage dividing circuit is applied to the input terminal of the control unit. This can prevent an operation error of the control unit or the like, which is caused by application of an overvoltage to the input terminal.

(78) Further features of the present invention will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a schematic block diagram of an exemplary arrangement of an aerosol inhalation device according to an embodiment of the present invention;

(2) FIG. 1B is a schematic block diagram of another exemplary arrangement of the aerosol inhalation device according to the embodiment of the present invention;

(3) FIG. 2A is a circuit diagram showing the circuit arrangement of an exemplary control device for the aerosol inhalation device according to the embodiment of the present invention;

(4) FIG. 2B is a circuit diagram showing the circuit arrangement of the exemplary control device for the aerosol inhalation device according to the embodiment of the present invention;

(5) FIG. 2C is a circuit diagram showing the circuit arrangement of another exemplary control device for the aerosol inhalation device according to the embodiment of the present invention;

(6) FIG. 2D is a circuit diagram showing the circuit arrangement of another exemplary control device for the aerosol inhalation device according to the embodiment of the present invention;

(7) FIG. 3A is a circuit diagram showing some circuits of the exemplary control device when a load is attached;

(8) FIG. 3B is a circuit diagram showing some circuits of the exemplary control device when a load is detached;

(9) FIG. 3C is a circuit diagram showing some circuits of the exemplary control device when a load is detached, for which a second shunt resistor is assumed to be absent;

(10) FIG. 3D is a circuit diagram showing some circuits of the exemplary control device, for which a diode electrically connected to an operational amplifier is assumed to be absent;

(11) FIG. 3E is a circuit diagram showing an equivalent circuit of a constituent element of some circuits of the exemplary control device;

(12) FIGS. 4A, 4A-2 and 4B show a flowchart of exemplary main processing for measuring the temperature of a load and judging exhaustion or shortage of an aerosol source; and

(13) FIG. 5 is a flowchart of exemplary processing for supporting main processing.

DESCRIPTION OF THE EMBODIMENTS

(14) An embodiment of the present invention will now be described in detail with reference to the accompanying drawings. Note that the embodiment of the present invention includes an electronic cigarette, a heated tobacco product, and a nebulizer, but is not limited to these. The embodiment of the present invention can include various aerosol inhalation devices configured to generate an aerosol to be inhaled by a user.

1 Outline of Aerosol Inhalation Device

(15) FIG. 1A is a schematic block diagram of the arrangement of an aerosol inhalation device **100A** according to an embodiment of the present invention. FIG. 1A schematically and conceptionally shows components provided in the aerosol inhalation device **100A** but does not show the strict

arrangement, shapes, dimensions, positional relationship, and the like of the components and the aerosol inhalation device **100A**.

(16) As shown in FIG. **1A**, the aerosol inhalation device **100A** includes a first member **102** (to be referred to as a “main body **102**” hereinafter), and a second member **104A** (to be referred to as a “cartridge **104A**” hereinafter). As shown in FIG. **1A**, as an example, the main body **102** may include a control unit **106**, a notification unit **108**, a power supply **110**, a sensor **112**, and a memory **114**. The aerosol inhalation device **100A** may include sensors such as a flow velocity sensor, a flow rate sensor, a pressure sensor, a voltage sensor, a current sensor, and a temperature sensor. In the present invention, these will also collectively be referred to as the “sensor **112**”. The main body **102** may also include a circuit **134** to be described later. As an example, the cartridge **104A** may include a storage unit **116A**, an atomization unit **118A**, an air intake channel **120**, an aerosol channel **121**, a mouthpiece portion **122**, a holding portion **130**, and a load **132**. Some of the components included in the main body **102** may be included in the cartridge **104A**. Some of the components included in the cartridge **104A** may be included in the main body **102**. The cartridge **104A** may be configured to be detachable from the main body **102**. Alternatively, all the components included in the main body **102** and the cartridge **104A** may be included in a single housing in place of the main body **102** and the cartridge **104A**.

(17) The storage unit **116A** may be formed as a tank to store an aerosol source. In this case, the aerosol source is, for example, a polyhydric alcohol such as glycerol or propylene, a liquid such as water, or a liquid mixture thereof. If the aerosol inhalation device **100A** is an electronic cigarette, the aerosol source in the storage unit **116A** may contain a component that discharges a flavor component when heated. The holding portion **130** holds the aerosol source supplied from the storage unit **116A** at a position where the load **132** can heat. For example, the holding portion **130** is made of a fibrous or porous material and holds an aerosol source as a liquid in gaps between fibers or in the pores of the porous material. As the above-described fibrous or porous material, for example, cotton, glass fiber, a tobacco raw material, or the like can be used. If the aerosol inhalation device **100A** is a medical inhalation device such as a nebulizer, the aerosol source may also contain a drug to be inhaled by a patient. As another example, the storage unit **116A** may include a component capable of replenishing the consumed aerosol source. Alternatively, the storage unit **116A** may be configured such that the storage unit **116A** itself can be exchanged when the aerosol source is consumed. The aerosol source is not limited to a liquid and may be a solid. If the aerosol source is a solid, the storage unit **116A** may be a hollow container.

(18) The atomization unit **118A** is configured to atomize the aerosol source and generate an aerosol. If an inhalation operation or another operation by the user is detected by the sensor **112**, the atomization unit **118A** generates an aerosol. For example, the holding portion **130** is provided to connect the storage unit **116A** and the atomization unit **118A**. In this case, a part of the holding portion **130** communicates with the inside of the storage unit **116A** and contacts the aerosol source. Another part of the holding portion **130** extends to the atomization unit **118A**. Note that the other part of the holding portion **130** extending to the atomization unit **118A** may be stored in the atomization unit **118A**, or may communicate with the inside of the storage unit **116A** via the atomization unit **118A**. The aerosol source is carried from the storage unit **116A** to the atomization unit **118A** by the capillary effect of the holding portion **130**. As an example, the atomization unit **118A** includes a heater including the load **132** electrically connected to the power supply **110**. Note that in the present application, “electrically connected” may indicate a state in which electricity can be transported between two elements electrically connected to each other. If electricity can be transported between two elements, the method can be either wired or wireless. Hence, a third element, for example, a resistor may be included between the two electrically connected elements or not. The heater is arranged in contact with or in proximity of the holding portion **130**. If an inhalation operation or another operation by the user is detected by the sensor **112**, the control unit **106** controls power supply to the heater of the atomization unit **118A** and heats the aerosol source

carried via the holding portion **130**, thereby atomizing the aerosol source. The air intake channel **120** is connected to the atomization unit **118A**, and the air intake channel **120** communicates with the outside of the aerosol inhalation device **100A**. The aerosol generated by the atomization unit **118A** is mixed with air taken via the air intake channel **120**. The fluid mixture of the aerosol and air is sent to the aerosol channel **121**, as indicated by an arrow **124**. The aerosol channel **121** has a tubular structure configured to transport the fluid mixture of air and the aerosol generated by the atomization unit **118A** to the mouthpiece portion **122**.

(19) The mouthpiece portion **122** is located at the end of the aerosol channel **121** and configured to open the aerosol channel **121** to the outside of the aerosol inhalation device **100A**. The user holds the mouthpiece portion **122** in the mouth and inhales, thereby taking the air containing the aerosol into the oral cavity.

(20) The notification unit **108** may include a light emitting element such as an LED, a display, a speaker, a vibrator, and the like. The notification unit **108** is configured to make some notification to the user as needed by light emission, display, utterance, vibration, or the like.

(21) Note that the cartridge **104A** can be formed as an outer tube, and one or both of the air intake channel **120** and the aerosol channel **121** can be formed as an inner tube arranged in the outer tube. The load **132** can be arranged in the air intake channel **120** or the aerosol channel **121** that is the inner tube. The storage unit **116A** can be arranged or formed between the cartridge **104A** that is the outer tube and the air intake channel **120** or the aerosol channel **121** that is the inner tube.

(22) The power supply **110** supplies power to the components such as the notification unit **108**, the sensor **112**, the memory **114**, the load **132**, and the circuit **134** in the aerosol inhalation device **100A**. The power supply **110** may be a primary battery or a secondary battery that can be charged by connecting to an external power supply via a predetermined port (not shown) of the aerosol inhalation device **100A**. Only the power supply **110** may be detachable from the main body **102** or the aerosol inhalation device **100A**, or may be exchangeable with a new power supply **110**. In addition, the power supply **110** may be exchangeable with a new power supply **110** by exchanging the whole main body **102** with a new main body **102**. As an example, the power supply **110** may be formed by a lithium ion secondary battery, a nickel hydrogen secondary battery, a lithium ion capacitor, or the like. The power supply **110** that is the secondary battery may include a temperature sensor configured to detect the temperature of the battery.

(23) The sensor **112** may include one or a plurality of sensors used to acquire the value of a voltage applied to the whole or a specific part of the circuit **134**, the value of a current flowing to the whole or a specific part of the circuit **134**, a value associated with the electrical resistance value of the load **132** or a value associated with the temperature, and the like. The sensor **112** may be incorporated in the circuit **134**. The function of the sensor **112** may be incorporated in the control unit **106**. The sensor **112** may also include at least one of a pressure sensor that detects a variation in the pressure in one or both of the air intake channel **120** and the aerosol channel **121**, a flow velocity sensor that detects a flow velocity, and a flow rate sensor that detects a flow rate. The sensor **112** may also include a weight sensor that detects the weight of a component such as the storage unit **116A**. The sensor **112** may also be configured to count the number of puffs by the user who uses the aerosol inhalation device **100A**. The sensor **112** may also be configured to integrate the time of energization to the atomization unit **118A**. The sensor **112** may also be configured to detect the liquid level in the storage unit **116A**. The sensor **112** may also be configured to obtain or detect the SOC (State Of Charge), current integrated value, voltage, and the like of the power supply **110**. The SOC may be obtained by the current integration method (coulomb counting method), the SOC-OCV (Open Circuit Voltage) method, or the like. The sensor **112** may also include the above-described temperature sensor in the power supply **110**. The sensor **112** may be able to detect an operation for an operation button that can be operated by the user.

(24) The control unit **106** can be an electronic circuit module formed as a microprocessor or a microcomputer, for example, an MPC. The control unit **106** may be configured to control the

operation of the aerosol inhalation device **100A** in accordance with computer executable instructions stored in the memory **114**. The memory **114** is a storage medium such as a ROM, a RAM, or a flash memory. In addition to the computer executable instructions as described above, the memory **114** may store setting data necessary for control of the aerosol inhalation device **100A**. For example, the memory **114** may store various data such as a control method (a form such as light emission, utterance, or vibration) of the notification unit **108**, values obtained and/or detected by the sensor **112**, and the heating history of the atomization unit **118A**. The control unit **106** reads out data from the memory **114** as needed and uses it to control the aerosol inhalation device **100A**, and stores data in the memory **114** as needed. Note that the memory **114** may be included in the control unit **106**.

(25) FIG. **1B** is a schematic block diagram of the arrangement of an aerosol inhalation device **100B** according to the embodiment of the present invention.

(26) As shown in FIG. **1B**, the aerosol inhalation device **100B** has an arrangement similar to the aerosol inhalation device **100A** shown in FIG. **1A**. However, the arrangement of the second member **104B** (to be referred to as an “aerosol generating article **104B**” or “stick **104B**” hereinafter) is different from the arrangement of the second member **104A**. As an example, the aerosol generating article **104B** may include an aerosol base material **116B**, an atomization unit **118B**, the air intake channel **120**, the aerosol channel **121**, and the mouthpiece portion **122**. Some of the components included in the main body **102** may be included in the aerosol generating article **104B**. Some of the components included in the aerosol generating article **104B** may be included in the main body **102**. The aerosol generating article **104B** may be insertable/removable into/from the main body **102**. Alternatively, all the components included in the main body **102** and the aerosol generating article **104B** may be included in a single housing in place of the main body **102** and the aerosol generating article **104B**.

(27) The aerosol base material **116B** may be formed as a solid carrying an aerosol source. As in the storage unit **116A** shown in FIG. **1A**, the aerosol source may be, for example, a polyhydric alcohol such as glycerol or propylene, a liquid such as water, or a liquid mixture thereof. The aerosol source in the aerosol base material **116B** may contain a tobacco raw material or an extract derived from a tobacco raw material, which discharges a flavor component when heated. Note that the aerosol base material **116B** itself may be made of a tobacco raw material. If the aerosol inhalation device **100B** is a medical inhalation device such as a nebulizer, the aerosol source may also contain a drug to be inhaled by a patient. The aerosol base material **116B** may be configured such that the aerosol base material **116B** itself can be exchanged when the aerosol source is consumed. The aerosol source is not limited to a liquid and may be a solid.

(28) The atomization unit **118B** is configured to atomize the aerosol source and generate an aerosol. If an inhalation operation or another operation by the user is detected by the sensor **112**, the atomization unit **118B** generates an aerosol. The atomization unit **118B** includes a heater (not shown) including a load electrically connected to the power supply **110**. If an inhalation operation or another operation by the user is detected, the control unit **106** controls power supply to the heater of the atomization unit **118B** and heats the aerosol source carried in the aerosol base material **116B**, thereby atomizing the aerosol source. The air intake channel **120** is connected to the atomization unit **118B**, and the air intake channel **120** communicates with the outside of the aerosol inhalation device **100B**. The aerosol generated by the atomization unit **118B** is mixed with air taken via the air intake channel **120**. The fluid mixture of the aerosol and air is sent to the aerosol channel **121**, as indicated by the arrow **124**. The aerosol channel **121** has a tubular structure configured to transport the fluid mixture of air and the aerosol generated by the atomization unit **118B** to the mouthpiece portion **122**.

(29) The control unit **106** is configured to control the aerosol inhalation devices **100A** and **100B** (to be collectively referred to as the “aerosol inhalation device **100**” hereinafter) according to the embodiment of the present invention by various methods.

2 Outline of Control Device for Aerosol Inhalation Device

(30) 2-1 First Control Device

(31) 2-1-1 Circuit Arrangement

(32) FIGS. 2A and 2B are circuit diagrams showing the schematic circuit arrangement of a control device **200A** for the aerosol inhalation device **100** according to the embodiment of the present invention. Reference numerals used to mention the elements of the control device **200A** are added in FIG. 2A, and reference symbols and the like concerning voltages and electrical resistance values used in the description are added in FIG. 2B. Note that of the notification unit **108**, the power supply **110**, the sensor **112**, the memory **114**, the control unit **106**, and the circuit **134**, the control device **200A** may include elements that are not illustrated.

(33) The control device **200A** includes the power supply **110**, the MCU that is an example of the control unit **106**, a sensor **112A** and a sensor **112C**, the load **132** (to be also referred to as a “heater resistor” hereinafter, and its electrical resistance value is represented by $R_{\text{sub.HTR}}$), a contact resistor **254** (its electrical resistance value is represented by $R_{\text{sub.CONNECTOR}}$) of the load **132** in a case in which the load **132** is detachable from the control device **200A**, a connection terminal **256** and a connection terminal **258** of the load **132** to circuits, a first circuit **202**, a second circuit **204**, a switch Q1 including a first field effect transistor (FET) **206**, a converter **208** and a regulator **242**, a switch Q2 including a second FET **210**, a first shunt resistor **212** and a second shunt resistor **252** (their electrical resistance values will be represented by $R_{\text{sub.shunt1}}$ and $R_{\text{sub.shunt2}}$, respectively, hereinafter), an operational amplifier **262**, a resistor **264** and a resistor **266** which are electrically connected to the inverting input terminal of the operational amplifier **262**, a diode **268**, a resistor **272** electrically connected to the output terminal of the operational amplifier **262**, and a capacitor **274** electrically connected to the resistor **272**. As for the diode **268**, the anode may electrically be connected to the noninverting input terminal of the operational amplifier **262**, and the cathode may electrically be connected to the power supply terminal of the operational amplifier **262** (more specifically, it can be a positive-side power supply terminal in a positive-side power supply terminal and a negative-side power supply terminal whose applied voltages are generally referred to as $V_{\text{sub.cc}}$ and $V_{\text{sub.EE}}$, respectively, and this will also apply to the power supply terminal of the operational amplifier **262** hereinafter). In other words, the potential at the cathode of the diode **268** may equal the potential at the power supply terminal of the operational amplifier **262**.

(34) In the control device **200A**, $V_{\text{sub.out}}$ corresponds to the output voltage of the converter **208**, $V_{\text{sub.sample}}$ corresponds to a voltage applied to the noninverting input terminal of the operational amplifier **262**, $V_{\text{sub.ref}}$ corresponds to a voltage applied to the inverting input terminal of the operational amplifier **262**, $V_{\text{sub.analog}}$ corresponds to a voltage according to the voltage of the output terminal of the operational amplifier **262**, which is a voltage applied to the control unit **106**, $V_{\text{sub.op-amp}}$ corresponds to a voltage applied to the power supply terminal of the operational amplifier **262**, that is, the power supply voltage of the operational amplifier **262**, and $V_{\text{sub.MCU}}$ corresponds to the output voltage of the regulator **242**, which is a voltage applied to the power supply terminal of the control unit **106**, that is, the power supply voltage of the control unit **106**. In the control device **200A**, $V_{\text{sub.op-amp}}$ equals $V_{\text{sub.MCU}}$.

(35) The electrical resistance value of the load **132** changes in accordance with the temperature. In other words, the load **132** can include a PTC heater. Since the electrical resistance value of the load **132** changes in accordance with the temperature, it can be said that the temperature of the load **132** and the electrical resistance value of the load **132** have a correlation. The load **132** is detachably electrically connected to the circuits of the control device **200A** via the connection terminal **256** and the connection terminal **258**. The load **132** may be included in the control device **200A** or not.

(36) The first shunt resistor **212** is electrically connected in series with the load **132** and has the known electrical resistance value $R_{\text{sub.shunt1}}$. The electrical resistance value $R_{\text{sub.shunt1}}$ of the first shunt resistor **212** can be almost or completely invariable with respect to the temperature. The

first shunt resistor **212** has an electrical resistance value larger than that of the load **132**. The second shunt resistor **252** can have the same characteristic as the first shunt resistor **212**, but is not limited to this. Note that the second shunt resistor **252** is electrically connected in series with the first shunt resistor **212**. In addition, the second shunt resistor **252** is electrically connected in parallel with the load **132**.

(37) The sensor **112A** and the sensor **112C** form a part of the sensor **112**. The sensor **112A** and the sensor **112C** may be omitted in accordance with the embodiment.

(38) The first FET **206** included in the switch **Q1** and the second FET **210** included in the switch **Q2** each play a role of a switch that opens/closes an electrical circuit. It would be obvious for those skilled in the art that as the switch, not only an FET but also various elements such as an IGBT and a contactor can be used to form the switches **Q1** and **Q2**. In addition, the switches **Q1** and **Q2** preferably have the same characteristic, but may not. Hence, the FETs, IGBTs, contactors, or the like used as the switches **Q1** and **Q2** preferably have the same characteristic, but may not. Note that if elements having the same characteristic are employed as the switches **Q1** and **Q2**, the procurement cost for each of the switches **Q1** and **Q2** can be reduced. This makes it possible to manufacture the control device **200A** at a lower cost.

(39) The converter **208** can be a DC/DC converter. The converter **208** is, for example, a switching regulator, and can include an FET **214**, a diode **216**, an inductor **218**, and a capacitor **220**. The control unit **106** may control the converter **208** such that the converter **208** converts the output voltage of the power supply **110**, and the converted output voltage becomes $V_{sub.out}$. Here, the converter **208** is preferably configured to output a predetermined voltage under the control of the control unit **106** during the time when at least the switch **Q2** is in an ON state. In addition, the converter **208** may be configured to output a predetermined voltage under the control of the control unit **106** during the time when the switch **Q1** is in an ON state as well. Note that in this case, the voltage output by the converter **208** need not strictly be constant. If the target voltage of the converter **208** is maintained constant for a predetermined period, it can be said that the converter **208** is configured to output a predetermined voltage. Note that the predetermined voltage output by the converter **208** under the control of the control unit **106** during the ON state of the switch **Q1** and the predetermined voltage output by the converter **208** under the control of the control unit **106** during the ON state of the switch **Q2** may be equal or different. If these are different, the predetermined voltage output by the converter **208** under the control of the control unit **106** during the ON state of the switch **Q1** may be higher or lower than the predetermined voltage output by the converter **208** under the control of the control unit **106** during the ON state of the switch **Q2**.

According to this arrangement, since the voltages and other parameters are stable, the remaining aerosol amount detection accuracy improves. Furthermore, when a switching regulator is used as the converter **208**, a loss generated when a voltage input to the converter **208** is converted into a predetermined voltage can be made small. This can generate a larger amount of aerosol by one charge while improving the remaining aerosol amount detection accuracy.

(40) The converter **208** may be configured to directly apply the output voltage of the power supply **110** to the first circuit **202** under the control of the control unit **106** during the time when only the switch **Q1** is in the ON state. This form may be implemented by the control unit **106** controlling the switching regulator **208** in a direct connection mode in which the switching operation stops. Note that the converter **208** is not an essential component but may be omitted. The converter **208** may be of a step-down type shown in FIG. 2A, or may be of a boost type or a step-down/boost type.

(41) Note that the control of the converter **208** may be done by another control unit other than the control unit **106**. The other control unit may be provided in the converter **208**. In this case, a value detected by the sensor **112C** is input at least to the other control unit. Note that in this case as well, a value detected by the sensor **112C** may be input to the control unit **106**.

(42) The regulator **242** can be, for example, a linear regulator and, more particularly, an LDO (Low Drop-Out Regulator). The regulator **242** is electrically connected to the power supply terminal of

the control unit **106**, and generates the voltage $V_{sub.MCU}$ used to drive the control unit **106**.

(43) The voltage $V_{sub.MCU}$ is a voltage used to drive the control unit **106** and can therefore be a relatively low voltage. On the other hand, the voltage $V_{sub.out}$ is associated with a voltage applied to the load **132**, and is preferably a relatively high voltage to improve the atomization efficiency. Hence, in general, the voltage $V_{sub.out}$ is higher than the voltage $V_{sub.MCU}$. For this reason, in the control device **200A**, the voltage $V_{sub.out}$ is higher than the voltage $V_{sub.op-amp}$ equal to the voltage $V_{sub.MCU}$. Hence, the control device **200A** is formed by at least a first region (for example, a region indicated by **290**, although it is not necessarily limited) where the maximum voltage is $V_{sub.out}$ and a second region (for example, a region indicated by **292**, although it is not necessarily limited) where the maximum voltage is $V_{sub.MCU}$. As described above, the maximum voltage $V_{sub.MCU}$ in the second region is lower than the maximum voltage $V_{sub.out}$ in the first region.

(44) Note that unless a boosting device exists in each region, the maximum voltage equals the voltage applied to the entire region. That is, as described above, it can be said that the voltage $V_{sub.MCU}$ applied to the second region is lower than the voltage $V_{sub.out}$ applied to the first region.

(45) The circuit **134** shown in FIGS. **1A** and **1B** electrically connects the power supply **110** and the load **132**, and can include the first circuit **202** and the second circuit **204**. The first circuit **202** and the second circuit **204** are electrically connected in parallel between the power supply **110** and the load **132**. The first circuit **202** can include the switch **Q1**. The second circuit **204** can include the switch **Q2** and the first shunt resistor **212**. Hence, the first circuit **202** can have an electrical resistance value smaller than that of the second circuit **204**. The circuit from the first circuit **202** to the load **132** forms at least a part of an aerosol generation circuit.

(46) As indicated by dotted arrows in FIGS. **2A** and **2B**, the control unit **106** can control the switch **Q1**, the switch **Q2**, and the like, and can acquire values detected by the sensor **112A** and the sensor **112C**. The control unit **106** may be configured to cause the first circuit **202** to function by switching the switch **Q1** from an OFF state to an ON state and cause the second circuit **204** to function by switching the switch **Q2** from an OFF state to an ON state. The control unit **106** may be configured to cause the first circuit **202** and the second circuit **204** to alternately function by alternately switching the switches **Q1** and **Q2**.

(47) The first circuit **202** is mainly used to atomize the aerosol source. When the switch **Q1** is switched to the ON state, and the first circuit **202** functions, power is supplied to the load **132**, and the load **132** is heated. By heating the load **132**, the aerosol source held by the holding portion **130** in the atomization unit **118A** (in the aerosol inhalation device **100B** shown in FIG. **1B**, the aerosol source carried by the aerosol base material **116B**) is atomized, and an aerosol is generated.

(48) The second circuit **204** is used to acquire, for example, the value of a voltage according to a voltage applied to the load **132**.

(49) The operational amplifier **262** is used to form a voltage sensor that forms a part of the sensor **112**. In the control device **200A**, the operational amplifier **262** forms a part of an amplification circuit. Hence, the voltage $V_{sub.analog}$ according to the voltage $V_{sub.sample}$ (exactly, according to the difference between the voltage $V_{sub.sample}$ and the voltage $V_{sub.ref}$) is applied to the control unit **106**. $V_{sub.analog}$ may not be the difference itself between the voltage $V_{sub.sample}$ and the voltage $V_{sub.ref}$, and may be a voltage obtained by amplifying the difference between the voltage $V_{sub.sample}$ and the voltage $V_{sub.ref}$. Note that the power supply terminal (not shown) of the operational amplifier **262** may electrically be connected to ground. Additionally, in the control device **200A**, the element electrically connected to the noninverting input terminal of the operational amplifier **262** and the element electrically connected to the inverting input terminal may be reversed. Hence, in some cases, the diode **268** is electrically connected to the inverting input terminal of the operational amplifier **262**.

(50) 2-1-2 Second Shunt Resistor **252**

(51) The second shunt resistor **252** is used to stabilize the voltage V.sub.sample and the voltage V.sub.analog according to it when the load **132** is detached from the aerosol inhalation device **100** and reliably detect the detachment of the load **132**, as will be described below in detail.

(52) FIGS. **3A** and **3B** are circuit diagram showing some circuits of the control device **200A** when the load **132** is attached or detached. In the following description, the switch **Q1** is assumed to be in the OFF state, and the switch **Q2** is assumed to be in the ON state.

(53) When the load **132** is attached, the voltage V.sub.sample applied to the noninverting input terminal of the operational amplifier **262** is a voltage obtained by dividing the voltage V.sub.out by the first shunt resistor **212** and a combined resistor **340** (its electrical resistance value is represented by R') of the second shunt resistor **252**, the resistor **254**, and the load **132**. That is,

$$(54) \quad V_{\text{sample}} = \frac{R'}{R_{\text{shunt1}} + R'} \cdot \text{Math. } V_{\text{out}} \quad (1)$$

Here,

$$(55) \quad \frac{1}{R'} = \frac{1}{R_{\text{shunt2}}} + \frac{1}{R_{\text{connector}} + R_{\text{HTR}}} \quad (2) \quad \frac{1}{R'} = \frac{R_{\text{shunt2}} + R_{\text{connector}} + R_{\text{HTR}}}{R_{\text{shunt2}} \cdot \text{Math. } (R_{\text{connector}} + R_{\text{HTR}})} R' = \frac{R_{\text{shunt2}} \cdot \text{Math. } (R_{\text{connector}} + R_{\text{HTR}})}{R_{\text{shunt2}} + R_{\text{connector}} + R_{\text{HTR}}}$$

(56) On the other hand, when the load **132** is detached, the voltage V.sub.sample applied to the noninverting input terminal of the operational amplifier **262** is a voltage obtained by dividing the voltage V.sub.out by the first shunt resistor **212** and the second shunt resistor **252**. That is,

$$(57) \quad V_{\text{sample}} = \frac{R_{\text{shunt2}}}{R_{\text{shunt1}} + R_{\text{shunt2}}} \cdot \text{Math. } V_{\text{out}} \quad (3)$$

(58) As described above, the first shunt resistor **212** and the second shunt resistor **252** have electrical resistance values sufficiently larger than that of the resistor **254** or the load **132**. Hence, since R' is obviously different from R.sub.shunt2, based on equations (2), the voltage V.sub.sample changes between a state in which the load **132** is attached and a state in which the load **132** is detached, as can be seen from equations (1) and (3). Hence, the voltage V.sub.analog according to the voltage V.sub.sample also changes between the state in which the load **132** is attached and the state in which the load **132** is detached. This allows the control unit **106** to detect detachment of the load **132** based on the applied voltage.

(59) Here, the circuits of the control device **200A** in which the second shunt resistor **252** is assumed to be absent, as shown in FIG. **3C**, will be examined. When the load **132** is detached, a path that starts from the power supply **110**, passes through the first shunt resistor **212**, and returns to the power supply **110** again does not form a closed circuit, and the value of the voltage V.sub.analog becomes unstable.

(60) Hence, the accuracy in detecting the detachment of the load **132** can greatly be improved by providing the second shunt resistor **252**.

(61) 2-1-3 Diode **268**

(62) The diode **268** is used to prevent an overvoltage from being applied to the noninverting input terminal of the operational amplifier **262**, as will be described below in detail.

(63) FIG. **3D** is a circuit diagram showing some circuits of the control device **200A**, for which the diode **268** is assumed to be absent.

(64) First, a case in which the switch **Q1** is in the ON state, and the switch **Q2** is in the OFF state in the circuit shown in FIG. **3D** will be examined. As is apparent from FIG. **2B**, the voltage V.sub.sample equals the voltage V.sub.out. Hence,

$$(65) \quad V_{\text{sample}} = V_{\text{out}} \quad (4)$$

Additionally, as described above, since

$$(66) \quad V_{\text{op-amp}} < V_{\text{out}} \quad (5)$$

we obtain, from inequalities (4) and (5)

$$(67) \quad V_{\text{op-amp}} < V_{\text{sample}} \quad (6)$$

(68) This represents that the voltage V.sub.sample applied to the noninverting input terminal of the operational amplifier **262** becomes higher than the power supply voltage V.sub.op-amp of the

operational amplifier **262**. In this case, an overvoltage is applied to the noninverting input terminal of the operational amplifier **262**, and the operational amplifier **262** may cause an operation error. (69) A case in which the switch Q1 is in the OFF state, and the switch Q2 is in the ON state in the circuit shown in FIG. 3D will be examined next. An electrical resistance value R of a combined resistor **342** of the first shunt resistor **212** and the combined resistor **340** (its electrical resistance value is represented by R') of the second shunt resistor **252**, the resistor **254**, and the load **132** is given by

$$(70) \quad R = R_{\text{shunt1}} + R' \quad (7)$$

(71) Since the electrical resistance value R' of the combined resistor **340** is represented by equations (2), the electrical resistance value R of the combined resistor **342** is given by

$$(72) \quad R = R_{\text{shunt1}} + \frac{R_{\text{shunt2}} \cdot \text{Math.} (R_{\text{connector}} + R_{\text{HTR}})}{R_{\text{shunt2}} + R_{\text{connector}} + R_{\text{HTR}}} \quad (8)$$

Letting I be the current flowing to the first shunt resistor **212**, I is given by

$$(73) \quad I = V_{\text{out}} / R \quad (9)$$

Since the voltage drop in the first shunt resistor **212** is I.Math.R.sub.shunt1, the voltage V.sub.sample can be represented by

$$(74) \quad \begin{aligned} V_{\text{sample}} &= V_{\text{out}} - I \cdot \text{Math.} R_{\text{shunt1}} \\ &= V_{\text{out}} - \frac{V_{\text{out}}}{R} \cdot \text{Math.} R_{\text{shunt1}} \\ &= V_{\text{out}} - V_{\text{out}} \cdot \text{Math.} R_{\text{shunt1}} \cdot \text{Math.} \left\{ R_{\text{shunt1}} + \frac{R_{\text{shunt2}} \cdot \text{Math.} (R_{\text{connector}} + R_{\text{HTR}})}{R_{\text{shunt2}} + R_{\text{connector}} + R_{\text{HTR}}} \right\}^{-1} \quad (10) \\ &= V_{\text{out}} - V_{\text{out}} \cdot \text{Math.} \left\{ 1 + \frac{R_{\text{shunt2}} \cdot \text{Math.} (R_{\text{connector}} + R_{\text{HTR}})}{R_{\text{shunt1}} \cdot \text{Math.} (R_{\text{shunt2}} + R_{\text{connector}} + R_{\text{HTR}})} \right\}^{-1} \end{aligned}$$

(75) According to equation (10), if R.sub.shunt1 and R.sub.shunt2 are much larger than R.sub.connector and R.sub.HTR, the voltage V.sub.sample applied to the noninverting input terminal of the operational amplifier **262** has a sufficiently small value. Hence, V sample does not become higher than the power supply voltage V.sub.op-amp of the operational amplifier **262**. Otherwise, however, the voltage V.sub.sample applied to the noninverting input terminal of the operational amplifier **262** becomes higher than the power supply voltage V.sub.op-amp of the operational amplifier **262**. That is, even if the switch Q1 is in the OFF state, and the switch Q2 is in the ON state, the voltage V.sub.sample applied to the noninverting input terminal of the operational amplifier **262** may be excessive.

(76) FIG. 3E shows an equivalent circuit **350** concerning the diode **268** in the control device **200A**, which is configured to solve the problem of the excessive voltage.

(77) Reference numeral **352** corresponds to the noninverting input terminal of the operational amplifier **262**, and **354** corresponds to the substantial resistance (its electrical resistance value will be represented by R.sub.op-amp hereinafter) of the operational amplifier **262**. Reference numerals **362** and **364** correspond to voltage sources that generate V.sub.op-amp and V.sub.sample, respectively.

(78) If the voltage V.sub.op-amp is almost higher than the voltage V.sub.sample, no forward current flows to the diode **268**. Hence, the current flows only in a direction indicated by **372**. On the other hand, if the voltage V.sub.sample is almost higher than the voltage V.sub.op-amp, a forward current flows to the diode **268**. Here, if the electrical resistance value R.sub.op-amp of the resistor **354** is sufficiently larger than the electrical resistance value of the diode **268** in the forward direction, most of a current having an overvoltage flows to the diode **268**.

(79) If an excessively large current flows to the diode **268**, the backflow preventing function of the diode **268** lowers, and a current generated by the voltage V.sub.op-amp may flow through the load **132**. Hence, the current flowing to the diode **268** needs to be smaller than the maximum value depending on the diode **268**. A condition to prevent the current flowing to the diode **268** from exceeding its maximum value will be examined below.

(80) In the equivalent circuit **350**, let I_{F_upper} be the maximum forward current permitted by the diode **268**, and R_{F_F} be the electrical resistance value of the diode **268** in the forward direction. If the following condition is satisfied, the current flowing to the diode **268** never exceeds its maximum value.

$$(81) \quad V_{sample} < I_{F_upper} \cdot R_F + V_{op-amp} \quad (11)$$

Here, since the voltage $V_{sub.sample}$ does not exceed the voltage $V_{sub.out}$, as described above, after all, if the following condition is satisfied, the current flowing to the diode **268** never exceeds its maximum value, and the backflow preventing function of the diode **268** does not lower.

$$(82) \quad V_{out} < I_{F_upper} \cdot R_F + V_{op-amp} \quad (12)$$

(83) As described above, the smaller the value of $V_{sub.out}$ is, the smaller the forward current of the diode **268** is. As a result, the diode **268** can be protected. However, in the ON state of the switch **Q1**, $V_{sub.out}$ is applied to the load **132** for generating the aerosol. Hence, if $V_{sub.out}$ is made too low, the load **132** may be unable to generate the aerosol.

(84) Hence, $V_{sub.out}$ needs to simultaneously satisfy the above-described condition of inequality (12) and a condition that the load **132** can generate the aerosol by the voltage. It was found as a result of earnest examinations of the present inventors that if the load **132** has an electrical resistance value of about 2.4Ω at room temperature, and $V_{sub.out}$ is about 3.3 to 3.7 V, the above-described two conditions are simultaneously satisfied. It was also found that if the load **132** has an electrical resistance value of about 1.0Ω at room temperature, and $V_{sub.out}$ is about 2.1 to 2.5 V, the above-described two conditions are simultaneously satisfied.

(85) In the above-described embodiment, a circuit arrangement that makes the input voltage $V_{sub.sample}$ of the operational amplifier **262** equal to or lower than the power supply voltage $V_{sub.op-amp}$ of the operational amplifier **262** has been described. The same problem can occur when a comparator is used in place of the operational amplifier **262**. Even in this case, the input voltage of the comparator can be made equal to or lower than the power supply voltage of the comparator by using the diode **268** whose anode is connected to the input terminal of the comparator.

(86) 2-2 Second Control Device

(87) 2-2-1 Circuit Arrangement

(88) FIGS. **2C** and **2D** are circuit diagrams showing the schematic circuit arrangement of a control device **200B** for the aerosol inhalation device **100** according to another embodiment of the present invention. Reference numerals used to mention the elements of the control device **200B** are added in FIG. **2C**, and reference symbols and the like concerning voltages and electrical resistance values used in the description are added in FIG. **2D**. The control device **200A** and the control device **200B** have some common components, and the same reference numerals denote the same components in FIGS. **2A** and **2B** and FIGS. **2C** and **2D**. Differences of the control device **200B** from the control device **200A** will be described below.

(89) In the control device **200B**, the diode **268** does not exist, and the voltage $V_{sub.out}$ is applied to the power supply terminal of the operational amplifier **262**. Hence, in the control device **200B**, the voltage $V_{sub.op-amp}$ applied to the power supply terminal of the operational amplifier **262** equals $V_{sub.out}$.

(90) Also, the control device **200B** includes a voltage dividing circuit **286** formed by a first resistor **282** and a second resistor **284** (their electrical resistance values will be represented by $R_{sub.div1}$ and $R_{sub.div2}$, respectively, hereinafter). The voltage dividing circuit **286** divides the voltage $V_{sub.analog}$ and outputs a voltage $V'_{sub.analog}$.

(91) Further, in the control device **200B**, the voltage $V'_{sub.analog}$ is input to the control unit **106**.

(92) 2-2-2 Voltage Dividing Circuit **286**

(93) In the control device **200B**, if the voltage dividing circuit **286** does not exist, the voltage $V_{sub.analog}$ is applied to the control unit **106**. Since the power supply voltage of the operational

amplifier **262** in the control device **200B** is $V_{sub.out}$, the voltage $V_{sub.analog}$ according to the output voltage of the operational amplifier **262** may also reach $V_{sub.out}$. On the other hand, since the power supply voltage of the control unit **106** is $V_{sub.MCU}$, and in general, $V_{sub.MCU} < V_{sub.out}$, as described above, the voltage $V_{sub.analog}$ higher than $V_{sub.MCU}$ that is the power supply voltage is applied to the control unit **106**, and an overvoltage may occur.

(94) The voltage dividing circuit **286** is configured to prevent the above-described problem. In the control device **200B**, the relationship between the voltage $V_{sub.analog}$ and the voltage $V'_{sub.analog}$ applied to the control unit **106** is represented by

$$(95) \quad V'_{analog} = \frac{R_{div2}}{R_{div1} + R_{div2}} \cdot V_{analog} \quad (13)$$

(96) To avoid input of the overvoltage to the control unit **106**, the voltage $V'_{sub.analog}$ input to the control unit **106** needs to be equal to or lower than the power supply voltage $V_{sub.MCU}$. That is,

$$(97) \quad \begin{aligned} V_{MCU} &\geq V'_{analog} \\ &\geq \frac{R_{div2}}{R_{div1} + R_{div2}} \cdot V_{analog} \end{aligned} \quad (14)$$

(98) Let $A_{sub.d}$ be the amplification factor of the amplification circuit formed by the operational amplifier **262** in the control device **200B**. Then, we obtain

$$(99) \quad V_{analog} = A_d \cdot (V_{sample} - V_{ref}) \quad (15)$$

Since the voltage $V_{sub.sample}$ does not exceed the voltage $V_{sub.out}$,

$$(100) \quad V_{analog} \leq A_d \cdot (V_{out} - V_{ref}) \quad (16)$$

(101) Based on inequalities (14) and (16), if the voltage $V_{sub.MCU}$ and the voltage $V_{sub.out}$ satisfy the following relationship, the above-described problem of the overvoltage does not occur.

$$(102) \quad V_{MCU} \geq \frac{R_{div2}}{R_{div1} + R_{div2}} \cdot V_{analog} \quad (17) \quad V_{MCU} \geq \frac{R_{div2}}{R_{div1} + R_{div2}} \cdot A_d \cdot (V_{out} - V_{ref})$$

$$\frac{R_{div1} + R_{div2}}{R_{div2}} \cdot \frac{1}{A_d} \cdot V_{MCU} \geq V_{out} - V_{ref} \quad V_{out} - V_{ref} \leq \frac{R_{div1} + R_{div2}}{R_{div2}} \cdot \frac{1}{A_d} \cdot V_{MCU} \quad (18)$$

$$V_{out} \leq \frac{R_{div1} + R_{div2}}{R_{div2}} \cdot \frac{1}{A_d} \cdot V_{MCU} + V_{ref}$$

(103) Hence, application of the overvoltage to the control unit **106** can be prevented by performing one or both of adjusting the converter **208** to output $V_{sub.out}$ satisfying inequalities (18) and setting the electrical resistance values $R_{sub.div1}$ and $R_{sub.div2}$ of the first resistor **282** and the second resistor **284** of the voltage dividing circuit **286** to satisfy inequalities (18). From another viewpoint, according to inequalities (18), the upper limit value of the output voltage $V_{sub.out}$ of the converter **208** in a case in which the electrical resistance values $R_{sub.div1}$ and $R_{sub.div2}$ of the first resistor **282** and the second resistor **284** of the voltage dividing circuit **286** are fixed is determined.

(104) As described above, the smaller the value of $V_{sub.out}$ is, the lower $V_{sub.analog}$ is. As a result, the control unit **106** can be protected. However, in the ON state of the switch **Q1**, $V_{sub.out}$ is applied to the load **132** for generating the aerosol. Hence, if $V_{sub.out}$ is made too low, the load **132** may be unable to generate the aerosol. Hence, $V_{sub.out}$ needs to simultaneously satisfy the above-described condition of inequalities (18) and a condition that the load **132** can generate the aerosol by the voltage. It was found as a result of earnest examinations of the present inventors that if the load **132** has an electrical resistance value of about 2.4Ω at room temperature, and $V_{sub.out}$ is about 3.3 to 3.7 V, the above-described two conditions are simultaneously satisfied. It was also found that if the load **132** has an electrical resistance value of about 1.0Ω at room temperature, and $V_{sub.out}$ is about 2.1 to 2.5 V, the above-described two conditions are simultaneously satisfied.

3 Operation of Control Device for Aerosol Inhalation Device

(105) 3-1 Main Processing

(106) FIGS. **4A**, **4A-2** and **4B** are flowcharts of exemplary main processing **400** for measuring the temperature of the load **132** and detecting the remaining amount of the aerosol source. Note that the main processing **400** is executed by the control device **200A** or the control unit **106** included in the

control device **200A**, and repeated during the operation of the aerosol inhalation device **100**.

(107) Step **410** represents a step of determining whether a first condition and a second condition are satisfied. Upon determining that the first condition and the second condition are satisfied, the process advances to step **420**. Otherwise, step **410** is repeated. The first condition and the second condition will be described later.

(108) In subsequent step **450** to be described later, a signal used to set the switch **Q1** in the ON state to atomize the aerosol source is transmitted. If at least one of the first condition and the second condition is not satisfied in step **410**, the process does not advance to step **450**. Hence, setting the switch **Q1** in the ON state is inhibited.

(109) Step **420** represents a step of determining whether an aerosol generation request is detected. Upon determining that the aerosol generation request is detected, the process advances to step **430**. Otherwise, step **420** is repeated.

(110) Upon detecting the start of inhalation by the user based on information obtained from, for example, a pressure sensor, a flow velocity sensor, a flow rate sensor, or the like, the control unit **106** can determine that the aerosol generation request is detected. More specifically, for example, if the output value of the pressure sensor, that is, the pressure is lower than a predetermined threshold, the control unit **106** can determine that the start of inhalation by the user is detected. Alternatively, for example, if the output value of the flow velocity sensor or the flow rate sensor, that is, the flow velocity or the flow rate is higher than a predetermined threshold, the control unit **106** can determine that the start of inhalation by the user is detected. Since this determination method enables aerosol generation according to user's feeling, the flow velocity sensor or the flow rate sensor is particularly preferable. Alternatively, if the output values of these sensors start continuously changing, the control unit **106** may determine that the start of inhalation by the user is detected. Otherwise, based on pressing of a button used to start aerosol generation, or the like, the control unit **106** may determine that the start of inhalation by the user is detected. Alternatively, based on both the information obtained from the pressure sensor, the flow velocity sensor, or the flow rate sensor and pressing of the button, the control unit **106** may determine that the start of inhalation by the user is detected.

(111) Step **430** represents a step of determining whether a counter is equal to or less than a predetermined counter threshold. If the counter is equal to or less than the predetermined counter threshold, the process advances to step **440**. Otherwise, the process advances to step **464** shown in FIG. **4B** to be described later.

(112) The predetermined counter threshold can be one or more predetermined values. The significance of step **430** will be described later.

(113) Step **435** represents a step of controlling the converter **208** to generate the voltage $V_{sub.out}$ set in step **534** to be described later. This step can be a step of performing nothing if the converter **208** is already outputting the voltage $V_{sub.out}$. Also, this step can be executed at any timing before power is supplied to the load **132**.

(114) Step **440** represents a step of determining whether a predetermined cooling time has elapsed from the stop of power feed to the load **132**. Upon determining that the predetermined cooling time has elapsed from the stop of power feed to the load **132**, the process advances to step **442**. Otherwise, the process advances to step **448**.

(115) Step **442** represents a step of transmitting a signal used to set the switch **Q2** in the ON state to acquire the electrical resistance value of the load **132**.

(116) Step **443** represents a step of acquiring the electrical resistance value of the load **132** as $R_{sub.2}$ based on the voltage $V_{sub.analog}$ or the voltage $V'_{sub.analog}$ applied to the control unit **106** and at least temporarily storing the electrical resistance value in the memory **114**. As will be described later, the electrical resistance value $R_{sub.2}$ is used to calibrate the correlation between the electrical resistance value and the temperature of the load **132**.

(117) The method of obtaining the electrical resistance value of the load **132** based on the voltage

V.sub.analog or the voltage V'.sub.analog is arbitrary. For example, the electrical resistance value of the load **132** is the function of the value of the voltage V.sub.analog or the voltage V'.sub.analog. Hence, the function is obtained in advance experimentally or by calculations, and in step **443**, the electrical resistance value of the load **132** can be acquired in accordance with the function.

(118) Note that since the voltage V.sub.analog or the voltage V'.sub.analog is the voltage according to the output of the operational amplifier **262**, the processing of acquiring the electrical resistance value of the load **132** based on the voltage V.sub.analog or the voltage V'.sub.analog is processing based on the output of the operational amplifier **262**.

(119) In addition, the processing of acquiring the electrical resistance value of the load **132** based on the voltage V.sub.analog or the voltage V'.sub.analog is obviously processing based on the voltage applied to the control unit **106**.

(120) Step **444** represents a step of transmitting a signal used to set the switch Q2 in the OFF state.

(121) Step **445** represents a step of substituting the electrical resistance value R.sub.2 acquired in immediately preceding step **443** into a variable representing a reference resistance value R.sub.ref to calibrate the correlation between the electrical resistance value and the temperature of the load **132** to acquire the temperature of the load **132** in step **455** to be described later.

(122) Step **448** represents a step of substituting one of the electrical resistance values R.sub.2 acquired in step **443** before the immediately preceding step or the value of an electrical resistance value R.sub.1 of the load **132** acquired at the time of exchange of the cartridge **104A** to be described later into the variable representing the reference resistance value R.sub.ref.

(123) Note that if the value of the variable representing the reference resistance value R.sub.ref during the operation of the aerosol inhalation device **100** is held, step **448** may be a step of substituting the value of the electrical resistance value R.sub.1 into the variable in a case in which the value of the variable representing the reference resistance value R.sub.ref is not set, and otherwise, performing nothing.

(124) Note that the substitution in step **445** and step **448** is processing for improving the accuracy of a temperature T.sub.HTR of the load **132** calculated by an equation used in step **455** to be described later, which represents the correlation between the electrical resistance value and the temperature of the load **132**, and is an example of processing of calibrating the correlation.

(125) Step **450** represents a step of transmitting a signal used to set the switch Q1 in the ON state to atomize the aerosol source.

(126) Step **451** represents a step of transmitting a signal used to set the switch Q2 in the ON state to acquire the electrical resistance value of the load **132**, and step **452** represents a step of transmitting a signal used to set the switch Q1 in the OFF state to accurately acquire the electrical resistance value of the load **132**. It does not matter which of steps **451** and **452** is executed first, and the steps may be executed simultaneously. If a delay from the transmission of the signal to the switch to the actual change of the state is taken into consideration, step **451** is preferably performed before step **452**.

(127) Step **453** represents a step of acquiring the electrical resistance value of the load **132** as R.sub.3 based on the voltage V.sub.analog or the voltage V'.sub.analog applied to the control unit **106**.

(128) Step **454** represents a step of transmitting a signal used to set the switch Q2 in the OFF state.

(129) Step **455** represents a step of acquiring the temperature T.sub.HTR of the load **132**. The value of the temperature T.sub.HTR of the load **132** can be obtain by

$$(130) \quad T_{\text{HTR}} = \frac{R_3 - R_{\text{ref}}}{\text{TCR}} + T_{\text{ref}} \quad (19)$$

where Tref is a predetermined reference temperature, and is preferably a temperature equal to the temperature of the load **132** when its electrical resistance value is R.sub.ref or a temperature close to the temperature. For example, when T.sub.ref is set to an ambient temperature (for example, room temperature or 25° C.) at which the aerosol inhalation device **100** is assumed to be used, the

temperature of the load **132** at the time of execution of step **443** or step **530** to be described later will be a temperature near $T_{sub.ref}$. $\alpha_{sub.TCR}$ is a known constant depending on the material of the load **132**.

(131) Note that as described above, the processing of acquiring $R_{sub.ref}$ and $R_{sub.3}$ in equation (19) is processing based on the voltage according to the output of the operational amplifier **262**. Hence, the processing of acquiring the temperature $T_{sub.HTR}$ of the load **132** in accordance with equation (19) is processing based on the voltage according to the output of the operational amplifier **262**. Similarly, the processing of acquiring the temperature $T_{sub.HTR}$ of the load **132** in accordance with equation (19) is also processing based on the voltage applied to the control unit **106**.

(132) Step **460** represents a step of adding the temperature $T_{sub.HTR}$ of the load **132** acquired in preceding step **455** to a list that is a data structure and making the temperature referable later. Note that the list is merely an example, and in step **460**, an arbitrary data structure capable of holding a plurality of data such as an array can be used. Note that unless it is judged in step **470** to be described later that the process should advance to step **480**, the process of step **460** is executed a plurality of times. If step **460** is executed a plurality of times, the temperature $T_{sub.HTR}$ of the load **132** in the data structure is not overwritten but added as many as the number of times of execution of the process of step **460**.

(133) Step **462** represents a step of determining whether the temperature $T_{sub.HTR}$ of the load **132** acquired in immediately preceding step **455** is lower than a predetermined first threshold. If the temperature $T_{sub.HTR}$ of the load **132** is lower than the first threshold, the process advances to step **470**. Otherwise, the process advances to step **464**.

(134) The first threshold is preferably a temperature at which exhaustion of the aerosol source is strongly suspected when the temperature of the load **132** exceeds it, and is, for example, 300° C.

(135) Step **464** represents a step of inhibiting the switch **Q1** from changing to the ON state.

(136) This step can be a step of setting a flag concerning the first condition in the memory **114**, and this flag can be canceled when the cartridge **104A** is exchanged. That is, in this example, the first condition is that the cartridge **104A** is exchanged. Until the flag is canceled, that is, until the cartridge **104A** is exchanged, the first condition is not satisfied, and the determination in step **410** is false. In other words, unless the flag is canceled, the determination in step **410** never becomes true.

(137) Note that steps **462** and **464** are processes based on the temperature $T_{sub.HTR}$ of the load **132**, and as described above, the processing of acquiring the temperature $T_{sub.HTR}$ of the load **132** in accordance with equation (19) is processing based on the voltage according to the output of the operational amplifier **262**. Hence, the processing concerning steps **462** and **464** can be processing based on the voltage according to the output of the operational amplifier **262**. Similarly, the processing concerning steps **462** and **464** can also be processing based on the voltage applied to the control unit **106**.

(138) Step **466** represents a step of making a predetermined notification on a UI (user interface) on the notification unit **108**.

(139) This notification can be a notification representing that the cartridge **104A** should be exchanged.

(140) Step **470** represents a step of determining whether the aerosol generation request has ended. Upon determining that the aerosol generation request has ended, the process advances to step **480**. Otherwise, the process returns to step **450**.

(141) Upon detecting the end of inhalation by the user based on information obtained from, for example, a pressure sensor, a flow velocity sensor, a flow rate sensor, or the like, the control unit **106** can determine that the aerosol generation request has ended. Here, for example, if the output value of the pressure sensor, that is, the pressure exceeds a predetermined threshold, the control unit **106** can determine that the end of inhalation by the user is detected, in other words, aerosol generation is not requested. Alternatively, for example, if the output value of the flow velocity

sensor or the flow rate sensor, that is, the flow velocity or the flow rate is lower than a predetermined threshold, the control unit **106** can determine that the end of inhalation by the user is detected, in other words, aerosol generation is not requested. Note that the threshold may be larger than the threshold in step **420**, equal to the threshold, or smaller than the threshold. Otherwise, based on release of a button used to start aerosol generation, or the like, the control unit **106** may determine that the end of inhalation by the user is detected, in other words, aerosol generation is not requested. Alternatively, if a predetermined condition that, for example, a predetermined time has elapsed from the pressing of the button used to start aerosol generation is satisfied, the control unit **106** may determine that the end of inhalation by the user is detected, in other words, aerosol generation is not requested.

(142) Step **480** represents a step of determining whether the maximum value in the list that holds one or more temperatures T.sub.HTR of the load **132** is smaller than a predetermined second threshold. If the maximum value is smaller than the second threshold, the process advances to step **488**. Otherwise, the process advances to step **482**.

(143) The second threshold is preferably a temperature at which exhaustion of the aerosol source is suspected when the temperature of the load **132** exceeds it, but at which there is also a possibility of temporary shortage of the aerosol source in the holding portion **130** due to, for example, a delay of aerosol source supply from the storage unit **116A**. Hence, the second threshold can be smaller than the first threshold and is, for example, 250° C.

(144) Step **482** represents a step of temporarily inhibiting the switch Q1 from changing to the ON state.

(145) This step can be a step of setting a flag concerning the second condition in the memory **114**, and this flag can be canceled when a predetermined time has elapsed from setting of the flag. That is, in this example, the second condition is that a predetermined time elapses from setting of the flag. Until the flag is canceled, that is, until a predetermined time has elapsed from setting of the flag, the second condition is not satisfied, and the determination in step **410** is temporarily false. In other words, unless the flag is canceled, the determination in step **410** never becomes true. Note that the predetermined time can be 10 sec or more, for example, 11 sec.

(146) Note that steps **480** and **482** are processes based on the temperature T.sub.HTR of the load **132**, and as described above, the processing of acquiring the temperature T.sub.HTR of the load **132** in accordance with equation (19) is processing based on the voltage according to the output of the operational amplifier **262**. Hence, the processing concerning steps **480** and **482** can be processing based on the voltage according to the output of the operational amplifier **262**. Similarly, the processing concerning steps **480** and **482** can also be processing based on the voltage applied to the control unit **106**.

(147) Step **484** represents a step of making a predetermined notification on the UI on the notification unit **108**.

(148) This notification can be a notification that promotes the user to wait for inhalation of the aerosol for a while.

(149) Step **486** represents a step of, for example, incrementing the counter by one.

(150) Step **488** represents a step of initializing the counter and the list. By this step, the counter can be 0, and the list can be made empty.

(151) In this embodiment, in step **480** after the aerosol generation request has ended, the temperature T.sub.HTR of the load **132** is compared with the second threshold. In place of this embodiment, the temperature T.sub.HTR of the load **132** may be compared with the second threshold before the aerosol generation request ends. In this case, if it is judged that the temperature T.sub.HTR of the load **132** is equal to or higher than the second threshold, comparison between the second threshold and the temperature T.sub.HTR of the load **132** need not be performed any more until the aerosol generation request ends.

(152) 3-2 Auxiliary Processing

(153) FIG. 5 is a flowchart of exemplary processing 500 for supporting the main processing 400. The auxiliary processing 500 can be executed simultaneously with or in parallel to the main processing 400.

(154) Step 510 represents a step of determining whether exchange of the cartridge 104A is detected. Since the cartridge 104A includes the load 132, as described in 2-1-2, the control unit 106 detects attachment/detachment of the load 132 based on the voltage $V_{sub.analog}$ (or the voltage $V'_{sub.analog}$), thereby detecting exchange of the cartridge 104A. If exchange of the cartridge 104A is detected, the process advances to step 520. Otherwise, step 510 is repeated.

(155) Step 520 represents a step of transmitting a signal used to set the switch Q2 in the ON state to acquire the electrical resistance value of the load 132.

(156) Step 530 represents a step of acquiring the electrical resistance value of the load 132 as $R_{sub.1}$ based on the voltage $V_{sub.analog}$ or the voltage $V'_{sub.analog}$ applied to the control unit 106 and at least temporarily storing it in the memory 114.

(157) Step 532 represents a step of identifying the type of the exchanged cartridge 104A. Identifying the type of the cartridge 104A includes specifying, based on the electrical resistance value $R_{sub.1}$ acquired in step 530, the type (including a material, for example, Nichrome or SUS) of the load 132 included in the cartridge.

(158) Step 534 represents a step of setting the output voltage $V_{sub.out}$ optimum for the load 132 based on the specified type of the load 132.

(159) In the embodiment as described in 2-1 in which the input voltage $V_{sub.sample}$ of the operational amplifier 262 can be higher than the power supply voltage $V_{sub.op-amp}$ of the operational amplifier 262, the optimum output voltage $V_{sub.out}$ is preferably a voltage that prevents an operation error caused by the overvoltage of the diode 268 and allows the load 132 to generate the aerosol. More specifically, the optimum output voltage $V_{sub.out}$ is preferably a voltage that prevents the forward voltage flowing to the diode 268 from exceeding an allowable value and allows the load 132 to generate the aerosol.

(160) In the embodiment as described in 2-2 in which the voltage $V_{sub.analog}$ input to the control unit 106 can be higher than the power supply voltage $V_{sub.MCU}$ of the control unit 106, the optimum output voltage $V_{sub.out}$ is preferably a voltage that prevents an operation error caused by the overvoltage of the control unit 106 and allows the load 132 to generate the aerosol. More specifically, it is preferably a voltage that makes the voltage $V_{sub.analog}$ or the voltage $V'_{sub.analog}$ input to the control unit 106 equal to or lower than the power supply voltage $V_{sub.MCU}$ of the control unit 106 and allows the load 132 to generate the aerosol.

(161) It was found as a result of earnest examinations of the present inventors that if the load 132 has an electrical resistance value of about 2.4Ω at room temperature, and $V_{sub.out}$ is about 3.3 to 3.7 V, conditions that the above-described optimum output voltage $V_{sub.out}$ should satisfy are satisfied. It was also found that if the load 132 has an electrical resistance value of about 1.0Ω at room temperature, and $V_{sub.out}$ is about 2.1 to 2.5 V, the above-described two conditions are simultaneously satisfied.

(162) In this way, the plurality of optimum output voltages $V_{sub.out}$ or the range of the optimum output voltages $V_{sub.out}$ is determined in accordance with the type of the load 132. The converter 208 can be configured to output the plurality of voltages or the voltages in the range in accordance with the type of the load 132.

(163) Note that steps 532 and 534 are processed based on $R_{sub.1}$, and processing of acquiring $R_{sub.1}$ is processing based on the voltage according to the output of the operational amplifier 262. Hence, the processing concerning steps 532 and 534 is processing based on the voltage according to the output of the operational amplifier 262. Similarly, the processing concerning steps 532 and 534 is also processing based on the voltage applied to the control unit 106.

(164) Step 540 represents a step of transmitting a signal used to set the switch Q2 in the OFF state.

(165) Step 550 represents a step of initializing the above-described counter and list used in the

main processing **400**.

(166) Steps **462**, **480**, and **510** described above can be implemented by converting an analog signal that is the voltage according to the output of the operational amplifier **262** to a digital value by an A/D converter provided in the control unit **106** and comparing it with a predetermined threshold. In place of this method, steps **462**, **480**, and **510** described above can also be implemented by using a comparator in place of the operational amplifier **262** and setting the reference voltage of the comparator to a value corresponding to a predetermined threshold. It would be obvious for those skilled in the art that even in a case in which a comparator is used, the comparator or the control unit can be protected from an overvoltage by the above-described diode **268** or the voltage dividing circuit **286**.

4 Conclusion

(167) While the embodiments of the present invention have been described above, it is to be understood that the embodiments are merely examples, and do not limit the scope of the present invention. It should be understood that modifications, additions, improvements, and the like of the embodiments can appropriately be made without departing from the spirit and scope of the present invention. The scope of the present invention should not be limited by any of the above-described embodiments, and should be limited only by the appended claims and their equivalents.

REFERENCE SIGNS LIST

(168) **100A**, **100B** . . . aerosol inhalation device **102** . . . main body **104A** . . . cartridge **104B** . . . aerosol generating article **106** . . . control unit (for example, MCU) **108** . . . notification unit **110** . . . power supply **112**, **112A**, **112C** . . . sensor **114** . . . memory **116A** . . . storage unit **116B** . . . aerosol base material **118A**, **118B** . . . atomization unit **120** . . . air intake channel **121** . . . aerosol channel **122** . . . mouthpiece portion **130** . . . holding portion **132** . . . load **134** . . . circuit **200A**, **200B** . . . control device for aerosol inhalation device **202** . . . first circuit **204** . . . second circuit **206**, **210**, **214** . . . FET **208** . . . converter (for example, switching regulator) **212** . . . first shunt resistor **216** . . . diode **218** . . . inductor **220** . . . capacitor **242** . . . regulator (for example, LDO) **252** . . . second shunt resistor **254** . . . resistor **256**, **258** . . . connection terminal **262** . . . operational amplifier **264**, **266** . . . resistor **268** . . . diode **272** . . . resistor **274** . . . capacitor **282**, **284** . . . resistor that forms voltage dividing circuit **286** . . . voltage dividing circuit **340**, **342** . . . combined resistor **350** . . . equivalent circuit **352** . . . noninverting input terminal of operational amplifier **354** . . . resistor provided in operational amplifier **362**, **364** . . . voltage source **372**, **374** . . . flow of current

Claims

1. An aerosol inhalation device, comprising: an operational amplifier including an output terminal configured to generate a voltage according to a power applied to a load configured to heat an aerosol source and having a correlation between a temperature and an electrical resistance value; a controller including an input terminal and configured to perform processing based on a voltage applied to the input terminal; an RC circuit connected to the output terminal of the operational amplifier and configured to generate a voltage supplied to the input terminal of the controller; a first switch electrically connected between the power supply and the load; and a first resistor electrically connected in series with the first switch, wherein a power supply voltage of the operational amplifier is higher than a power supply voltage of the controller, and an inverting input terminal of the operational amplifier is electrically connected to the aerosol generation circuit.
2. The device according to claim 1, wherein the RC circuit is configured such that the voltage applied to the input terminal of the controller is not more than a power supply voltage of the controller.
3. The device according to claim 1, wherein an amplification factor of the operational amplifier is set such that the voltage applied to the input terminal of the controller is not more than the power supply voltage of the controller.

4. The device according to claim 1, wherein the device is configured to removably receive an aerosol generating article comprising the load and an aerosol base material, and wherein the controller is configured to detect an exchange of the aerosol generating article based on output of the operational amplifier.
 5. The device according to claim 2, wherein the device is configured to removably receive an aerosol generating article comprising the load and an aerosol base material, and wherein the controller is configured to detect an exchange of the aerosol generating article based on output of the operational amplifier.
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